

Mengqi Shi

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EDUCATION

University of Washington

M.S. in Technology Innovation – Connected Devices. GPA [3.92/4.00]

Seattle, WA
Sep 2025 – Mar 2027 (expected)

University of Washington

B.A. in Communication, minors in Informatics and Data Science. GPA [3.83/4.00]

Seattle, WA
Sep 2021 – Jun 2025

RESEARCH INTERESTS

Accessibility, Responsible AI, Social Computing, Empirical HCI, Inclusive and Human-Centered Technologies

RESEARCH EXPERIENCE

AI4SG Lab

Research Assistant | Advisor: Yi-Chieh (EJ) Lee, Tianqi (Holly) Song

National University of Singapore

Apr 2025 – Present

Emotional Safety Boundaries and Controls in Older Adults' Use of AI Companions

- Initiated and led an independent qualitative study on how older adults establish emotional safety boundaries when using conversational AI for companionship, partnering with a community organization to recruit 18 adults aged 60+ for semi-structured interviews.
- Conducted the full qualitative research pipeline, including study design, interviewing, coding, and thematic analysis, to identify concerns around emotional reliance, self-disclosure, and agency in human-AI companionship.
- Translated findings into design implications for emotionally safer, context-aware, and user-controllable AI companions, resulting in a poster accepted at CHI EA '26 and a full paper accepted at DIS '26.

Peer Conversational Agents for Older Adults' AI Literacy

- Co-conducted semi-structured interviews with adults aged 50+ to investigate AI literacy needs, learning preferences, and perceptions of peer-like conversational agents as learning support.
- Contributed to rigorous inductive qualitative analysis through transcript verification, double-coding, reconciliation, and affinity diagramming, identifying key affective barriers including low confidence and disengagement.
- Helped derive four design guidelines for peer conversational agents supporting older adults' AI literacy; manuscript under review at IJHCS as third author.

ACE Lab

Research Assistant | Advisor: Jacob O. Wobbrock, James A. Fogarty, Mingyuan Zhong

University of Washington

Apr 2025 – Present

LLM- and Agent-Driven Accessibility Auditing and Repair for Web and Mobile Interfaces

- Develop and evaluate LLM- and agent-driven approaches for detecting, assessing, and repairing accessibility issues in web and mobile interfaces, grounded in WCAG 2.2.
- Build a dataset of web accessibility issues for system development and evaluation, including creating approximately 80% of the mock-up webpages used to represent diverse accessibility barriers.
- Contribute to system implementation and preliminary validation of accessibility auditing workflows; related work is in preparation for submission to ASSETS '26.

PlayBridge: Making Educational Games More Accessible for Children with Motor Impairments

- Investigate accessibility barriers in educational gameplay for children with motor impairments, focusing on input demands, interaction constraints, and participation challenges in play-based learning.
- Design, prototype, and iteratively refine physical and interactive systems that support more accessible and engaging gameplay experiences for children with diverse motor abilities.
- Translate accessibility needs into inclusive design directions for child-centered assistive gaming technologies through collaborative prototyping and evaluation.

Concept Inventories in Computing Education

- Conducted a structured literature review of concept inventory research in computing education, screening over 200 papers, extracting thousands of data points, and curating an analytic corpus of more than 100 studies across more than 10 dimensions.
- Synthesized and analyzed the literature to characterize methodological patterns, research trends, and gaps in the field; developed publication-quality visualizations in Matplotlib and Seaborn to support comparative analysis and evidence-driven interpretation.
- Co-authored a paper published at ICER '23, contributing to scholarly understanding of assessment design and validation practices in computing education research.

PUBLICATIONS

Peer Reviewed

- [1] **Designing with Tensions: Older Adults' Emotional Support-Seeking Under System-Level Constraints in Conversational AI**
Mengqi Shi, Tianqi Song, Zicheng Zhu, Yi-Chieh Lee | *DIS 2026 Full Paper* | Paper
- [2] **Taking Stock of Concept Inventories in Computing Education: A Systematic Literature Review**
Murtaza Ali, Sourojit Ghosh, Prerna Rao, Raveena Dhegaskar*, Sophia Jawort*, Alix Medler*, **Mengqi Shi***, Sayamindu Dasgupta | *ICER 2023 Full Paper* | Paper
**Equal contribution*

Lightly Reviewed

- [3] **When Humans Don't Feel Like an Option: Contextual Factors That Shape When Older Adults Turn to Conversational AI for Emotional Support**
Mengqi Shi, Tianqi Song, Zicheng Zhu, Yi-Chieh Lee | *CHI 2026 Poster* | Paper
- [4] **Relational Co-Adaptation in Emotionally Supportive AI: Tensions in Authentic Emotional Interaction**
Mengqi Shi | *CHI 2026 BiAlign Workshop Position Paper* | Paper

Under Review

- [5] **Learn AI with AI: Designing Conversational Agents to Support Older Adults in Learning about AI**
Eugene Tang KangJie, Tianqi Song, **Mengqi Shi**, Yi-Chieh Lee | *Under review for IJHCS*

WORK EXPERIENCE

Information School

Foundational Skills for Data Science Teaching Assistant | Advisor: Prof. Ott Toomet Dec 2024 – Jun 2025

- Supported instruction for an introductory data science course by leading lab sections, holding regular office hours, and tutoring students on foundational analytical concepts and coursework.
- Served as a liaison between students and the instructor through TA meetings, student support, and exam review sessions, helping clarify course expectations and address learning challenges.

Xiaomi Inc.

User Research Intern

Beijing, China

Jun 2024 – Sep 2024

- Led 50+ user interviews on mobile AI models, synthesizing qualitative findings into research reports that informed product refinement and experience design decisions.
- Curated and validated 3,000+ competitor labeling data points to support comparative analysis and internal evaluation of AI product performance.
- Developed data visualizations and supported the authorship of Xiaomi SU7's 2024 H1 in-car voice sentiment analysis report, examining patterns in public perception and user feedback.

SERVICE

